

Ayush Agarwal

23BCE10678 — ayush20039939@gmail.com — +91 9122941107
linkedin.com/in/ayush20039939 — github.com/Ayush99392003 — lexai-3fd1a.web.app

Experience

Docu3C Technologies

AI/ML Intern

- Designed and deployed multi-agent AI systems using MCP (Model Context Protocol), enabling automated tool-calling workflows for financial document processing
- Engineered LLM pipelines with structured outputs (Pydantic, Instructor) serving 3+ document intelligence use cases; reduced manual review time by an estimated 60%
- Architected real-time backend services with FastAPI, SSE, and WebSockets, handling concurrent async requests across multiple AI workflows
- Built full-stack validation interfaces (React, Streamlit) for AI output review; provisioned Azure ML GPU infrastructure, reducing iteration cycles from days to hours
- Enforced PEP8 standards, Ruff linting, and modular architecture across the codebase; managed dependencies with uv

Projects

LexAI – Production AI Platform

Cloud-Native Hybrid RAG

Deployed: lexai-3fd1a.web.app

- Single-handedly built and deployed a full-stack legal intelligence platform indexing 26,274 judgments (1.8GB corpus) with live production traffic
- Combined SQLite FTS5 (BM25), FAISS vector search (MPNet), PageRank graph ranking, and Reciprocal Rank Fusion into a unified retrieval engine
- Achieved 67.8 Recall@10 (+3.6% over baselines) with sub-1.1s query latency on 4Gi RAM / 2 CPU infrastructure
- Integrated Gemini 1.5 Flash for conversational querying; delivered async FastAPI services with REST + WebSocket endpoints
- Deployed React (Vite) frontend on Firebase Hosting with Google Sign-In auth; containerized backend on Google Cloud Run with Docker, autoscaling, and Cloud Build CI/CD

PolarBrief AI – Legal Argument Analyzer

Full-Stack LLM App

- Created document analysis tool that extracts, classifies (Pro/Con/Neutral), and scores legal arguments from unstructured PDFs
- Orchestrated LLMs via LangChain for summarization; combined TF-IDF with LLM outputs for weighted ranking
- Implemented hybrid OCR pipeline (Tesseract + pdfplumber) with Streamlit interface for report generation

Knowledge Graph Pipeline

NLP + Graph ML

- Constructed end-to-end KG extraction pipeline: OCR → chunking → entity-relation extraction → fuzzy resolution → validation
- Visualized canonical knowledge graphs using NetworkX and PyVis for interactive exploration

Technical Skills

- Languages:** Python, TypeScript, JavaScript, HTML/CSS
- AI/ML:** LLMs, RAG, NLP, Multi-Agent Systems, Prompt Engineering
- Agentic AI:** MCP, Tool Calling, Function Calling, Pydantic, Instructor
- Frameworks:** FastAPI, React, Vite, Node.js, LangChain, Scikit-learn
- Retrieval:** FAISS, BM25, SQLite FTS5, Vector DBs, PageRank, RRF
- Databases:** PostgreSQL, Firestore, SQLite, CosmosDB
- Cloud/DevOps:** GCP (Run, Build, Storage), Azure, Docker, Firebase, CI/CD
- APIs:** REST, WebSockets, SSE, AsyncIO, CORS, OAuth 2.0, Google Sign-In
- Tools:** Git, GitHub, Tesseract OCR, pdfplumber, NetworkX, PyVis
- Practices:** Agile, PEP8, Ruff, uv, Autoscaling, Health Checks, Testing

Education

B. Tech, CSE (Core)

VIT Bhopal University

CGPA: 8.52/10

Sep 2023 – Ongoing

Certifications

- Microsoft Certified: Azure Data Fundamentals – Microsoft (Jun 2025)
- Introduction to Machine Learning – NPTEL (May 2025)
- Introduction to Large Language Models – Google Cloud (Apr 2024)
- Introduction to Generative AI – Google Cloud (Apr 2024)
- Prompt Design in Vertex AI – Google Cloud (Jul 2024)

Achievements & Additional Information

- Top 25 team, Mahakumbh Hackathon
- Solved 125+ DSA problems on LeetCode
- Event Lead, Linpack Club, VIT Bhopal — Organized 5+ technical workshops (200+ participants)
- Represented university volleyball team (Thunders) at Aavahan sports fest, VIT Bhopal